

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

COURSE NAME: Fundamentals of Data Science

COURSE CODE: DSA0401

COURSE OUTCOME COVERED

CO1: Analyze the role of data science, facets of data, and the big data ecosystem for informed decision-making. (BL2)

QUESTIONS: 20

Total Marks:100

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I Bindhu. B (192524472) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Bindhu. B

Annexure A

Short Answer Test

1. A college wants to identify students who may fail in the final exam. How can Data Science help?
2. A hospital has patient records for many years. How can Data Science be applied?
3. An online shopping website wants to recommend products to customers. Which use of Data Science is applied here?
4. A bank wants to detect suspicious transactions. How can Data Science be useful?
5. A company wants to reduce customer loss. How can Data Science support this?
6. A school wants to improve student performance. Mention one benefit of Data Science in this case.
7. A supermarket wants to know which products sell more during festivals. How can Data Science help?
8. A transport department wants to reduce traffic congestion. How can Data Science be applied?
9. A farmer wants to improve crop yield. Give one use of Data Science.
10. A company has data in Excel tables, customer reviews, images, and videos. Which facet of data is shown here?
11. A weather app updates temperature every minute. Which facet of data is involved?
12. A university stores student details in rows and columns. What type of data structure is this?
13. Customer comments on social media are collected for analysis. What type of data is this?
14. A company collects a very large amount of daily user activity data from its mobile app. Which Big Data characteristic is applied?
15. A hospital dataset contains missing age values and incorrect phone numbers. Which data quality issue is present?
16. A business wants to convert customer data into useful sales decisions. Which Big Data characteristic is important?
17. A company stores large files in Hadoop Distributed File System. Which part of the Big Data ecosystem is used?
18. A company wants to process large datasets using parallel computation. Which Big Data tool can be used?
19. An organization wants to collect live sensor data from IoT devices. Which Big Data ecosystem component is suitable?
20. A company wants to visualize Big Data results for decision-makers. Which tool or method can be applied?

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

CO1 Assessment Tool-1

2) A hospital has patient records for many years. How can Data Science be applied?

Ans) Data Science can analyze patient records to identify disease patterns, predict health risks, and improve patient treatments.

4) A Bank wants to detect suspicious transactions. How can data Science be useful?

Ans) Data Science can analyze transaction patterns and detect unusual or fraudulent transactions.

6) A School wants to improve student performance. mention one benefit of Data Science in this case.

Ans) Data Science can identify students who need extra support and help improve their academic performance.

8) A transport department wants to reduce traffic congestion. How can data Science be applied?

Data Science can analyze traffic data is available in different formats such as tablets, text, images, and videos.

10) A company has data in Excel tables, customer reviews, images and videos. which facet of data is shown here?

ans) Variety - because the data is available in different formats such as tables, text, images and videos.

12) A university stores student details in rows and columns - what type of data structure is this?

If it is structured data, because the information is organized in rows and columns.

14) A company collects a very large amount of daily user activity data from its mobile app. which Big Data characteristic is applied?

Volume - because a very large amount of data is being generated and collected

1 business wants to convert customer data into useful sales decisions. which Big data characteristic is important?

value because the data is converted into useful information for making business decisions.

A company wants to process large datasets using parallel computation. which Big data tool can be used?

Apache spark can be used to process large datasets using parallel computation.

2) A company wants to visualize Big data results for decision makers, which tool or method can be applied?

Data visualization tool such as Tableau or Power BI can be used to present Big data results clearly for decision making.